Engineering Full-Duplex Voice Agents Around External Reasoning Models:
A Systems Survey and Reference Architecture

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Abstract

Full-duplex voice agents listen while speaking, handle overlap, and negotiate the conversational floor without forcing rigid turns. Recent surveys organize full-duplex spoken language models by synchronization strategy, model depth, interaction ontology, and learned representation. This report addresses a different engineering question: how can a duplex interaction layer retain an external, configurable reasoning and tool-using model as the agent brain? We synthesize 27 archived papers and three additional foundational/platform sources into a modular reference architecture. The architecture separates acoustic cleanup, target-speaker evidence, turn projection, interaction arbitration, asynchronous reasoning, speech rendering, and two consistency ledgers. We argue for a two-stage interruption protocol—reversible ducking followed by semantic commit—and for a hard side-effect boundary around speculative inference. We also propose an evaluation matrix that reports timing, arbitration, acoustic robustness, context consistency, and tool correctness separately. The result is a staged path from a cascaded local voice system to robust duplex behavior without replacing its configured language-model endpoint.

1 Introduction

Human conversation is not a sequence of perfectly separated recordings. Speakers pause, overlap, acknowledge, self-correct, and occasionally interrupt one another. Cross-linguistic evidence also shows that turn exchange is rapid but variable rather than governed by a fixed silence threshold (Stivers et al., 2009). A voice agent therefore needs more than voice activity detection (VAD): it must infer the role and intent of an acoustic event while its own speech is present.

Native full-duplex spoken language models such as Moshi, SyncLLM, SALM-Duplex, F-Actor, DuplexSLA, DuplexOmni, and JoyAI-Talker learn some or all of this behavior inside synchronized speech models (Défossez et al., 2024; Veluri et al., 2024; Hu et al., 2025; Züfle et al., 2026; Zhang et al., 2026; Huang et al., 2026; Bai et al., 2026). This line establishes an important behavioral ceiling, but it is not always a product fit. Coding and tool-using agents often need to preserve a configured reasoning endpoint, enterprise routing, model governance, and a mature action loop. Replacing that brain with a speech foundation model changes the product rather than merely adding voice.

This report surveys the complementary systems path: a real-time interaction layer around an external reasoning model. Its contributions are:

1. a signal-to-action decomposition connecting AEC, personalized speech evidence, semantic endpointing, duplex arbitration, tools, and TTS;

2. a reversible interruption protocol that separates audible responsiveness from irreversible conversational and tool state;
3. explicit heard-prefix and committed-action ledgers for consistency; and
4. a component-level evaluation program grounded in recent duplex benchmarks.

2 Scope and Relationship to Existing Surveys

The broad WavChat survey organizes spoken dialogue systems, representations, training, streaming, and evaluation across both cascaded and end-to-end models (Ji et al., 2024). Two dedicated full-duplex surveys then sharpen the space. Chen and Yu (2025) distinguish engineered from learned synchronization and group evaluation into temporal, behavioral, semantic, and acoustic pillars. Lu et al. (2026) introduce an L0–L3 architecture hierarchy, a $T \times I \times R$ interaction ontology, and an `IDLE/LISTEN/SPEAK/WAIT/DUAL` state machine.

Table 1 shows why another general taxonomy would be redundant. Our scope is instead the deployment boundary between a duplex voice layer and an external reasoning/tool agent. We include native models as behavioral references, but focus on components that can be measured, replaced, and integrated independently.

Table 1: Relationship to prior surveys. "External brain" means a separately configured language-model and tool runtime rather than the speech model's own reasoner.

Work	Primary organizing idea	Main distinction from this report
WavChat (Ji et al., 2024)	Cascaded vs. end-to-end spoken dialogue	Broader model, representation, and training survey
Chen and Yu (Chen and Yu, 2025)	Engineered vs. learned synchronization	Full-duplex model taxonomy and four-pillar evaluation
Lu et al. (Lu et al., 2026)	L0–L3 hierarchy, $T \times I \times R$, five-state machine	Architectural and interaction ontology across claimed duplex systems
This report	Replaceable interaction layer around an external brain	AEC, target-speaker gating, reversible interruption, ledgers, safe tools, and deployment order

3 Operational Definition and Signal Model

We use *full duplex* for a system that continuously accepts user audio while rendering assistant audio and can make interaction decisions from their overlap. This definition describes an interface capability, not proof that the system behaves well. A push-to-talk system is half duplex; a system that merely stops TTS whenever VAD fires is interruptible but behaviorally fragile; a capable duplex system distinguishes at least echo, noise, backchannel, other-addressed speech, hesitation, and substantive interruption.

Let u_t be near-end target-user speech, r_t the exact assistant render signal, h_t the room/device echo path, and v_t all other interference. The microphone observes

$$m_t = u_t + (h_t * r_t) + v_t, \tag{1}$$

where $*$ denotes convolution. The front end estimates the echo path and produces

$$e_t = m_t - (\hat{h}_t * r_t) \approx u_t + v_t. \tag{2}$$

This is why acoustic echo cancellation (AEC) belongs inside the system boundary. Apple voice processing and WebRTC AudioProcessing both use the render path as a reference rather than asking a language model to infer its own playback from a contaminated transcript (Apple Developer Documentation, 2026; WebRTC Project, 2026).

The interaction layer observes a feature vector

$$\boldsymbol{z}_t = [p_{\text{speech}}, p_{\text{target}}, \rho_{\text{echo}}, \boldsymbol{h}_t^{\text{audio}}, \boldsymbol{h}_t^{\text{text}}, q_t, \ell_t], \tag{3}$$

where q_t is dialogue state and ℓ_t is playback-ledger state. VAD is one feature, not the policy.

4 Architecture Families

4.1 Native synchronized speech models

dGSLM demonstrates that two streams of discrete speech units can learn turn-taking and overlap jointly without a text bottleneck (Nguyen et al., 2022). Voice Activity Projection (VAP) provides a related discriminative view by predicting both speakers' future activity rather than only current voice presence (Ekstedt and Skantze, 2022); later work demonstrates real-time operation and backchannel prediction (Inoue et al., 2024a,b).

Modern native systems place synchronized user and assistant streams on a shared clock. SyncLLM time-aligns fixed audio chunks (Veluri et al., 2024); Moshi combines user audio, assistant audio, and an assistant inner-monologue text stream (Défossez et al., 2024); SALM-Duplex fuses a streaming speech encoder with text and codec outputs (Hu et al., 2025). These designs reduce hand-built boundaries but couple dialogue policy, language reasoning, and speech generation.

F-Actor adds instruction control over voice and conversational behavior (Züfle et al., 2026); DuplexSLA adds a synchronized action channel (Zhang et al., 2026); JoyAI-Talker separates a Thinker, expressive Talker, and plug-in duplex gate (Bai et al., 2026). These advances inform interfaces even when their full models are not adopted.

4.2 Pluggable interaction controllers

FlexDuo wraps a half-duplex dialogue system with an audio window, context manager, state manager, and seven-action controller (Liao et al., 2025). A smaller semantic dialogue manager predicts four listen/speak control tokens after AEC, VAD, and ASR (Zhang et al., 2025). FireRedChat is the closest released production-shaped reference: personalized VAD (pVAD) rejects non-primary speakers, semantic end-of-turn detection decides completion, and a replaceable dialogue manager handles tools and context (Chen et al., 2025).

DuplexOmni generalizes this seam into a fast interaction layer and an asynchronous thinking layer (Huang et al., 2026). That split is the best match for an external reasoning endpoint: interaction never waits for deep reasoning, and obsolete thinking can be superseded when the user changes the task.

5 Reference Architecture

Figure 1 synthesizes the literature into a modular path. The front end removes known playback and estimates target-speaker evidence. A fast acoustic path reacts immediately; a streaming semantic path gathers partial text and prosody. The arbiter emits typed decisions. The external agent and TTS operate asynchronously, while ledgers retain what was heard and what actions were committed.

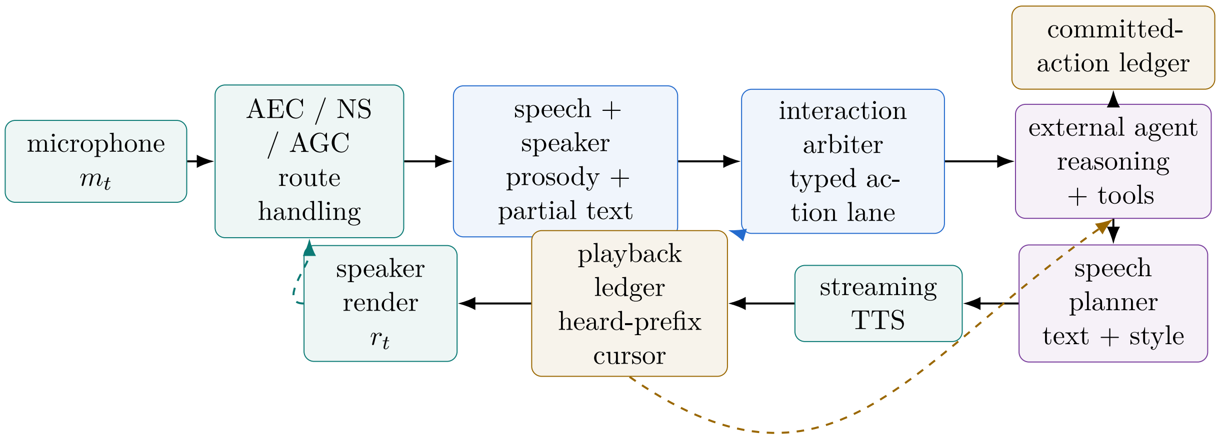

Figure 1: Reference architecture for a duplex voice layer around an external reasoning and tool agent. Black arrows show the primary data flow; the blue arrow controls playback; the dashed teal arrow supplies the exact render reference; and the dashed ochre arrow reconciles the heard prefix.

5.1 Target-speaker and interference gating

AEC answers whether audio resembles the assistant render. It does not answer whether residual speech belongs to the intended user. FireRedChat addresses the latter with pVAD (Chen et al., 2025). IRAF learns a causal reliability gate from a target-speaker embedding $\boldsymbol{s}$ and user-audio embedding $\boldsymbol{x}_t$ (Zhong et al., 2026):

$$g_t = 2\,\sigma(f_\psi(\boldsymbol{s}, \boldsymbol{x}_{\le t}))\,, \qquad \tilde{\boldsymbol{x}}_t = g_t \boldsymbol{x}_t. \tag{4}$$

For a modular system, p_{target} should remain visible alongside p_{speech} and echo residual. Collapsing them into one opaque score makes route changes and double-talk failures difficult to diagnose.

5.2 Turn projection and endpointing

VAP estimates future joint voice activity; textual models estimate syntactic and pragmatic completion. Their errors are complementary in conversational HRI (Skantze and Irfan, 2025). FastTurn combines streaming CTC text, acoustic representations, and a semantic model to distinguish complete, incomplete, backchannel, and wait events (Wang et al., 2026a). SpeculativeETD uses a cheap streaming detector and invokes a larger endpoint model only near a possible boundary (Ok et al., 2025).

Binary endpoint labels discard useful duration information. Next-Turn predicts

$$\tau_t = t_{\text{next onset}} - t, \tag{5}$$

which can be trained from timestamps and fused with endpoint probability (Tsoi et al., 2026). Endpoint Anticipation instead forecasts completion far enough ahead to prepare a short LLM/TTS continuation privately (Udupa et al., 2026). These are performance signals, not permission to commit user intent.

6 Interaction Arbitration and Reversible Interruption

Let Y_t be the latent event class (echo, noise, backchannel, other-addressed speech, hesitation, or substantive interruption) and $\mathcal{A}$ the controller actions. A calibrated policy minimizes asymmetric risk:

$$a_t^* = \arg\min_{a \in \mathcal{A}} \mathbb{E}[L(a, Y_t) \mid \boldsymbol{z}_{\leq t}] . \tag{6}$$

The loss should make audible talk-over expensive, but should also make destructive false cancellation more expensive than a brief duck. This motivates two phases:

1. **Reversible onset response.** Likely target speech immediately ducks or pauses playback. The agent request and remaining PCM stay resumable while 200–400 ms of evidence accumulates.
2. **Semantic commit.** A substantive interruption cancels future playback, drains stale capture, supersedes obsolete reasoning, and commits the new user turn. Echo, noise, backchannel, and other-addressed speech resume from the playback cursor.

This protocol makes reaction time depend on a fast acoustic path without giving that path authority over irreversible state. Session policy should also be configurable: F-Actor shows that backchannel and interruption behavior can be instruction-controlled rather than globally fixed (Züfle et al., 2026).

7 External Reasoning, Tools, and Consistency

The interaction layer needs a narrow contract with the external agent:

```
submit_committed_turn(transcript, timing, interruption_context)
supersede_response(response_id, heard_assistant_prefix)
receive_text_delta(response_id, text)
receive_tool_event(response_id, event)
```

The contract is deliberately text-and-event based. A custom endpoint can be replaced without teaching the acoustic controller about model vendors.

7.1 Heard-prefix ledger

If response text T_r has been synthesized but playback cursor $k_r(t)$ has only crossed a prefix, conversational history must contain

$$H_r(t) = T_r[0 : k_r(t)], \tag{7}$$

not the generated-but-unheard suffix. Otherwise the reasoning model believes the user heard facts or instructions that never reached the speaker. The playback ledger associates response IDs, text spans, PCM chunks, render timestamps, route changes, and cancellation state.

7.2 Committed-action ledger

Speech cancellation cannot undo a file edit or network side effect. Tool events therefore need a separate append-only ledger with proposed, authorized, started, committed, failed, and compensated states. DuplexSLA's distinct action lane supports this separation (Zhang et al., 2026). A predicted endpoint may initiate side-effect-free text/TTS speculation, but no mutating tool call may cross the commit boundary before the user turn is confirmed.

8 Speech Rendering and Expressivity

Native speech models preserve prosody because audio remains inside the model; cascaded systems often flatten intent into plain text. A modular compromise is a speech-plan object

$$p_i = (\mathrm{text}_i, \mathrm{intent}_i, \mathrm{rate}_i, \mathrm{pause}_i, \mathrm{emphasis}_i, \mathrm{affect}_i) \tag{8}$$

for every speakable clause. The external agent supplies semantics; a local planner assigns delivery; TTS renders incrementally. JoyAI-Talker's expressive Talker and F-Actor's promptable conversational behavior provide model-scale evidence for this interface (Bai et al., 2026; Züfle et al., 2026). Duplex correctness must not depend on any one TTS engine.

9 Evaluation

No scalar "duplex score" is sufficient. A system can appear highly responsive by cancelling on every noise burst. Evaluation should report the five groups in Table 2 separately.

Table 2: Component-level evaluation for modular duplex agents.

Group	Example metrics	Failure exposed
Acoustic	echo return loss enhancement, residual echo, double-talk recall, target-speaker confusion	self-barge and non-target activation
Timing	endpoint latency, stop latency, resume gap, audible response latency	sluggishness or premature response
Arbitration	interruption recall, false abandon, backchannel/background false stop	wrong conversational action
Consistency	heard-prefix accuracy, stale-frame count, superseded-response leakage	mismatch between heard and modeled history
Tools	argument accuracy, stale action rate, unauthorized speculative effects, compensation rate	real-world side effects from unstable turns

For AEC, a useful diagnostic is echo return loss enhancement

$$\mathrm{ERLE} = 10\log_{10}\frac{\mathbb{E}[d_t^2]}{\mathbb{E}[e_t^2]}, \tag{9}$$

where d_t is microphone echo before cancellation and e_t residual echo. ERLE must be paired with double-talk tests: suppressing both echo and the real user is not success.

Full-Duplex-Bench v1.5 isolates respond-versus-resume behavior under overlap (Lin et al., 2025a); v2 adds automated multi-turn examination (Lin et al., 2025b); v3 combines real disfluency with chained tool use (Lin et al., 2026). HumDial-FDBench contributes real dual-channel human conversations and a public challenge framework (Wang et al., 2026b). These external benchmarks should complement deterministic virtual-microphone tests with configurable room delay, gain, filtering, jitter, and overlap.

10 Implementation Roadmap

The literature supports the following order for a local modular system:

1. **Audio front end.** Add macOS voice processing with an exact render reference; retain WebRTC AudioProcessing as the cross-platform candidate. Log pre/post AEC, route changes, and underruns.

2. **Reversible playback.** Replace immediate energy-triggered cancellation with duck, classify, then resume or commit.

3. **Endpoint model.** Combine a small audio-native end-of-turn model with partial-text completion while preserving the silence fallback.

4. **Five-class arbiter.** Begin with calibrated rules for `HOLD_USER`, `TAKE_TURN`, `CONTINUE_AGENT`, `YIELD_AGENT`, and `IGNORE_INPUT`; train only after labeled feature traces exist.

5. **Ledgers and agent bridge.** Reconcile the heard prefix and committed tool effects on interruption and supersession.

6. **Safe anticipation.** Cache short speculative text/TTS output only after correctness is established; never speculatively mutate.

7. **Expressivity and native baselines.** Add speech planning, then compare against obtainable native systems without changing the external brain contract.

11 Open Problems

Robust target identity. Speaker embeddings drift across microphones, distance, illness, and noise. The field lacks a broadly validated target-speaker gate under simultaneous assistant render and near-end speech.

Training labels for intent under overlap. Public corpora contain far less annotated echo, other-addressed speech, backchannel, self-correction, and tool-state revision than clean turns. The recent surveys identify this realization gap (Chen and Yu, 2025; Lu et al., 2026).

Latency versus semantic authority. Endpoint anticipation can hide sequential delay, but premature semantic or tool commit is costly. Better calibrated uncertainty and explicit rollback domains are needed.

Cross-platform acoustic control. OS voice-processing graphs and WebRTC APM have different routing, device, and latency behavior. Reproducible duplex evaluation must include real devices, not only clean WAV files.

Expressivity without policy coupling. Native models jointly learn wording, timing, and voice. Modular systems need interfaces that preserve paralinguistic intent while keeping the agent, controller, and renderer independently replaceable.

12 Conclusion

The duplex literature does not imply that every voice product should train a native speech foundation model. It does imply that turn-taking is a continuous, multi-signal control problem. For agents whose value lies in an existing reasoning and tool loop, the strongest design is a deep interaction module: clean the signal with an exact render reference, retain target-speaker and semantic evidence separately, react reversibly, commit deliberately, and record both heard speech and irreversible actions. This architecture captures the interaction/thinking split emerging in recent models while preserving the configured external brain.

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